



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 5

1. Case Study Title: Implementation of JavaScript Array Operations Using Built-in Methods

Case Study Program Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">

  <title>Array Operations,Min/Max, Slice/Splice
[24070521207 SharvariWakde]</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background: #f2f2f2;
      text-align: center;
      padding: 40px;
    }

    .container {
      width: 500px;
      margin: auto;
      background: white;
      padding: 30px;
      border-radius: 12px;
      box-shadow: 0 0 10px gray;
    }

    input, select {
      width: 90%;
      padding: 10px;
      margin: 8px;
      font-size: 16px;
    }

    button {
      padding: 12px 25px;
      margin-top: 10px;
      background: #333;
      color: white;
      border: none;
      border-radius: 5px;
```



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```
<input type="number" id="sliceStart">

<label>Slice End:</label>
<input type="number" id="sliceEnd">
<br><br>
<button onclick="arrayOperations()">Perform
Operations</button>
<p id="output"></p>
<script src="shreya_1196_case5.js"></script>
</body>
</html>
```

```
function arrayOperations() {
    let arr = document.getElementById("arrayInput").value
        .split(",") .map(Number);
    let pushValue =
Number(document.getElementById("pushValue").value);
    let unshiftValue =
Number(document.getElementById("unshiftValue").value);

    let spliceIndex =
Number(document.getElementById("spliceIndex").value);
    let spliceValue =
Number(document.getElementById("spliceValue").value);

    let sliceStart =
Number(document.getElementById("sliceStart").value);
    let sliceEnd =
Number(document.getElementById("sliceEnd").value);
    let output = "";
    output += "Original Array: " + arr + "<br><br>";
    arr.push(pushValue);
    output += "After push: " + arr + "<br>";
```



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```
// pop()
arr.pop();
output += "After pop: " + arr + "<br>";

// unshift()
arr.unshift(unshiftValue);
output += "After unshift: " + arr + "<br>";

// shift()
arr.shift();
output += "After shift: " + arr + "<br>";

// splice()
arr.splice(spliceIndex, 1, spliceValue);
output += "After splice: " + arr + "<br>";

// slice()
let newArray = arr.slice(sliceStart, sliceEnd);
output += "After slice: " + newArray + "<br><br>";

// Maximum and Minimum
let maximum = Math.max(...arr);
let minimum = Math.min(...arr);

output += "Maximum: " + maximum + "<br>";
output += "Minimum: " + minimum;

document.getElementById("output").innerHTML = output;
}
```



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3. Output:

Array Operations, Min/Max, Slice

File C:/Users/HP/Desktop/24070521207_SharvariW(JS)/casestudy5.html

Array Operations, Min/Max, Splice/slice [24070521207_Sharvariwakde]

Enter numbers separated by commas:

What do you want to do?

Find Maximum

Submit

Maximum

Array: 10, 20, 30, 40, 50

Maximum value: 50

4. Result/Conclusion:

Thus, various JavaScript array operations were successfully performed and the results were displayed using user input.